

# Divide Whole Numbers by Fractions

Lesson 2-2

Name: \_\_\_\_\_

Date: \_\_\_\_\_

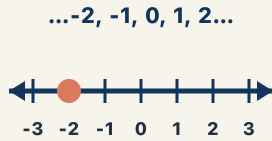
Class: \_\_\_\_\_

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## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



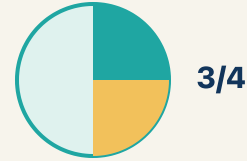
*4 is a whole number; as a fraction it is  $4/1$*

### Whole Number

Write the definition:

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*$1/2$  means 1 out of 2 equal parts — like cutting a sandwich in half and taking one piece*

### Fraction

Write the definition:

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$$3/4 \rightarrow 4/3$$

flip top and bottom

*The reciprocal of  $2/3$  is  $3/2$ . Flip the fraction to divide:  $6 \div 2/3 = 6 \times 3/2$*

### Reciprocal

Write the definition:

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$$3/4 \rightarrow 4/3$$

flip top and bottom

*$6 \div 1/3 \rightarrow$  Keep 6, Change  $\div$  to  $\times$ , Flip  $1/3$  to  $3/1 \rightarrow 6 \times 3 = 18$*

### Keep, Change, Flip

Write the definition:

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$$12 \div 3 = 4$$

*In  $4 \div 1/2 = 8$ , the quotient is 8*

## Quotient

Write the definition:

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

I can divide a whole number by a fraction by multiplying by the reciprocal.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Whole Number

Fraction

Reciprocal

Keep, Change, Flip

Quotient

1 A counting number like 0, 1, 2, or 3 with no fraction or decimal part is a \_\_\_\_\_.

2 A number that shows part of a whole, like  $\frac{3}{4}$ , is a \_\_\_\_\_.

3 The fraction you get by flipping a fraction upside down is its \_\_\_\_\_.

4 To divide by a fraction, I \_\_\_\_\_ : keep the first number, change  $\div$  to  $\times$ , and flip the second fraction.

5 The answer to a division problem is the \_\_\_\_\_.



### Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** What is  $6 \div \frac{1}{3}$ ?

- A. 18
- B. 2
- C.  $\frac{6}{3}$
- D. 3

**Step 1**  $6 \div \frac{1}{3} = 6 \times \frac{3}{1} = 18$ .

**Step 2** There are 18 thirds in 6 wholes.

 **Answer:** A. 18

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** What is  $3 \div \frac{1}{5}$ ?

- A. 15
- B.  $\frac{3}{5}$
- C.  $\frac{5}{3}$
- D. 8

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** What is  $10 \div \frac{1}{4}$ ?

- A. 40
- B. 2.5
- C. 14
- D.  $\frac{10}{4}$

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. There are 3 bags of trail mix. Each snack pouch needs  $\frac{2}{3}$  of a bag. How many full pouches can be filled, and is there a leftover?**

- A.  $4\frac{1}{2}$  pouches (4 full, half a pouch left)
- B. 4 pouches exactly
- C. 2 pouches
- D. 6 pouches

Show your work:

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**2. Detective Park bakes with 9 cups of batter. Each muffin pan holds  $\frac{3}{4}$  of a cup. How many pans can he fill?**

- A. 12
- B. 6
- C.  $27\frac{3}{4}$
- D. 9

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A pizza shop has 8 whole pizzas. Each serving is  $\frac{2}{3}$  of a pizza. How many servings can they make? Write and solve an equation, then explain why the answer is greater than 8.**

*Sentence starter: The equation is  $8 \div \frac{2}{3} = \underline{\hspace{1cm}}$ . Using Keep, Change, Flip:  $8 \times \frac{3}{2} = \underline{\hspace{1cm}}$ . The answer is greater than 8 because  $\underline{\hspace{1cm}}$ .*

Show your work:

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## Reflect — Exit Ticket

**What is  $9 \div \frac{1}{3}$ ?**

- A. 27
- B. 3
- C.  $\frac{9}{3}$
- D.  $\frac{1}{27}$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Whole Number
2. **Notes 2:** Fraction
3. **Notes 3:** Reciprocal
4. **Notes 4:** Keep, Change, Flip
5. **Notes 5:** Quotient
6. **We Try (notes):** A.  $15 - 3 \div 1/5 = 3 \times 5/1 = 15$ . *There are 15 fifths in 3 wholes.*
7. **You Try (notes):** A.  $40 - 10 \div 1/4 = 10 \times 4/1 = 40$ . *There are 40 quarter-size pieces in 10 wholes.*
8. **Try It 1:** A.  $4 \frac{1}{2}$  pouches (4 full, half a pouch left)  $- 3 \div 2/3 = 3 \times 3/2 = 9/2 = 4 \frac{1}{2}$ . *You can fill 4 full pouches with half a pouch of mix left over.*
9. **Try It 2:** A.  $12 - 9 \div 3/4 = 9 \times 4/3 = 36/3 = 12$  pans.
10. **Exit Ticket:** A.  $27 - 9 \div 1/3 = 9 \times 3/1 = 27$ . *There are 27 thirds in 9 wholes.*